

6 Skeletal muscle contains slow twitch and fast twitch muscle fibres.

(a) The table compares the percentage of slow twitch fibres in different types of athlete.

Type of athlete	Range of percentages of slow twitch fibres in muscle (%)
100m runner	12 to 30
1500m runner	30 to 65
10 000 m runner	60 to 95
marathon (42 200 m) runner	60 to 95

Explain the importance of slow twitch muscle fibres in these types of athlete.

(3)

[illegible]

(b) Name the basic unit of contraction in a skeletal muscle.

(1)

(c) The heart pumps oxygen rich blood to the tissues.

Cardiac output and cardiac index can show how effectively the heart is working.

- (i) A patient had a heart rate of $65 \text{ beats min}^{-1}$ and a stroke volume of 57 cm^3 per beat.

Calculate the cardiac output for this individual.

Use the equation:

$$\text{stroke volume} = \text{cardiac output} \div \text{heart rate}$$

Give your answer in $\text{dm}^3 \text{ min}^{-1}$.

(2)

Answer

The cardiac index can be calculated using the formula:

$$\text{cardiac index} = \frac{\text{stroke volume} \times \text{heart rate}}{\text{body surface area}}$$

The table shows data for two patients.

Patient	heart rate /bpm	stroke volume /dm ³	body surface area /m ²
A	79	0.057	1.54
B	79	0.075	2.39

(ii) The cardiac index for patient A was 2.92 dm³ min⁻¹ m⁻²

Which is the correct cardiac index for patient B?

(1)

- ☐ **A** 1.24
- ☐ **B** 2.48
- ☐ **C** 2.92
- ☐ **D** 5.21

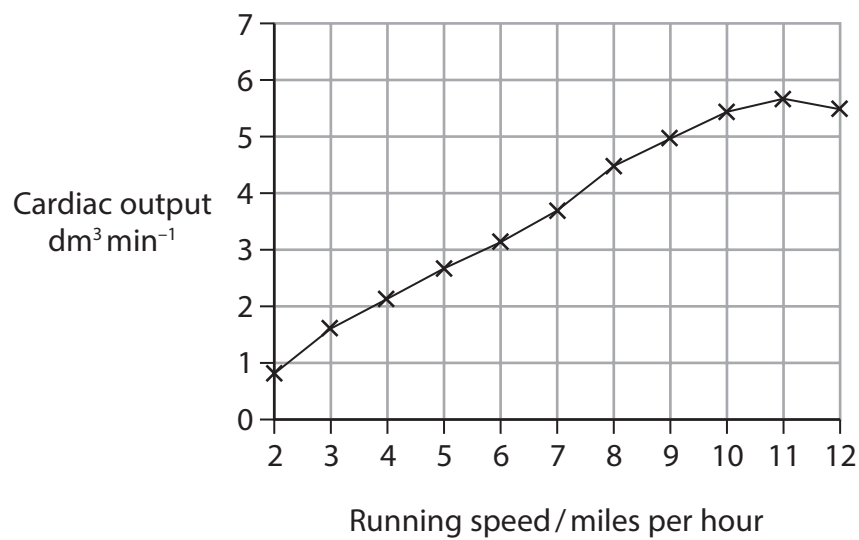
(iii) State the ratio of the body surface area of patient A to patient B.

(1)

Answer

(d) The cardiac output changes in an athlete as their running speed increases.

The graph shows cardiac output for an athlete running at different speeds.



(i) Comment on the effect of running speed on cardiac output.

(2)

(ii) Exercise increases the rate of respiration in muscles.

Explain how an increase in the rate of respiration causes an increase in the heart rate of an athlete.

(3)

(iii) Anaerobic respiration produces lactate.

Describe what happens to this lactate.

(2)

(Total for Question 6 = 15 marks)
